

Exam #1

Math 742 - Mathematical Statistics

2/11/2026

Name_____

Directions: The in-class portion of exam 1 is worth 50 points. The exam is split into 2 sections. Part I (problems 1-3) is 50 minutes. Part II (problems 4-5) is 25 minutes. You may have a hand held calculator for the arithmetic and a 8 1/2 x 11" sheet (one-sided) of notes.

Part I

Problem 1 (10 points)

Let X have pdf

$$f(x) = \begin{cases} 2x & : 0 < x < 1, \\ 0 & : \text{otherwise.} \end{cases}$$

- (a) Compute $\mathbb{E}[X]$ and $\mathbb{V}(X)$.
- (b) Let $Y = X^2$. Find the pdf of Y .
- (c) Verify Jensen's inequality for $\phi(x) = x^2$ by comparing $\mathbb{E}[X^2]$ and $(\mathbb{E}[X])^2$.

Problem 2 (10 points)

Let (X, Y) have joint density

$$f(x, y) = \begin{cases} 2 & : 0 < y < x < 1, \\ 0 & : \text{otherwise.} \end{cases}$$

- (a) Find the marginal density of X and compute $\mathbb{E}[X]$.
- (b) Find the conditional density $f_{Y|X}(y | x)$.
- (c) Compute $\text{Cov}(X, Y)$.

Problem 3 (10 points)

Let $X \sim \text{Exponential}(\lambda)$ and define $Y = \log X$.

- (a) Find the pdf of Y .
- (b) Does Y belong to a location family? Explain.
- (c) Find $\mathbb{E}[e^Y]$.